

Carltiliana Staidum
Software Engineer

(985) 856-8425 | carltianastaidum1@gmail.com | <https://github.com/Cece209?tab=repositories>

Software Engineering – Computer Science

Louisiana State University
Baton Rouge, Louisiana

Objective:

“Motivated Software Engineer/Computer Science graduate with a solid foundation in software development and programming languages. Seeking to apply coding skills and problem-solving abilities in a dynamic engineering team.”

Work Experience

Software Engineering 2022-2024, Louisiana State University

Front-End Developer – “Building Dreams” Website

Software Systems Developer | 2024 Louisiana State University

- I developed, designed, and implemented a full-stack website using HTML, CSS, and SQL.
- I integrated front-end and back-end functionality, including a database for user interactions.
- I applied software quality testing principles to verify performance and reliability.
- I built a React-based web application with AWS Amplify backend integration for authentication and data management.
- I developed a responsive front-end using React and Next.js, strengthening modern website development skills.

Programmer – Hotel Booking System

Object-Oriented Programming Developer | 2023 Louisiana State University

- I developed a C++ hotel reservation system with filters for city, price, and amenities.
- I applied object-oriented programming concepts to create modular, reusable code.
- I tested and debugged the application to ensure it met the course project requirements.

Programmer – Code Testing and Enhancement

Software Quality and Testing Developer | 2024 Louisiana State University

- I applied Junit, symmetric testing, concatenated testing, and other testing methodologies to validate functionality, ensure reliability, and improve maintainability across diverse projects.
- I designed and implemented original software solutions, developed custom code alongside writing comprehensive test suites to ensure correctness and robustness.
- I debugged and refined multiple codebases, enhancing performance and long-term sustainability.

Database Developer – NFL Stats Logger & PC Parts Picker Website

Database Management Systems Developer, 2024 Louisiana State University

- I built SQL databases to store and query NFL statistics and PC component information.
- I designed a front-end interface for the PC Parts Picker website to enhance user interaction.
- I optimized queries for data accuracy and retrieval efficiency.

Game Developer – Java, JavaScript, and C++ Games

Programming Digital Media & Game Design Developer, 2022-2024 Louisiana State University

- I implemented and created interactive games, including Math Game, *Red Light, Green Light*, *Hoverboard Hunter*, and a dynamic ping pong game.
- I developed a C++ maze generation game emphasizing logic and pathfinding.
- I applied game engine tools, testing, and debugging to improve performance.
- *Hoverboard Hunter* in Unity, I demonstrated transferable scripting/crawler-style logic by programming automated responses to specific key or controller inputs
- Configured reverse proxy settings to handle routing, managing backend connections, and securing database connections.
- Explored creating tools and systems within Unity that empower the players and creators with new content-building experiences.
- I integrated Arduino Uno hardware with software logic to manage real-time user input and event-driven gameplay.

Technical Skills

Data Structures: Arrays, Stacks, Queues, Linked Lists, Trees

Algorithms: Sorting, Searching, Recursion

Big O Notation: For understanding performance

Programming & Tools: C++, Java, JavaScript, Python, SQL(PostgreSQL, MySQL), NoSQL(MongoDB, Redis) Ruby, OCaml, CTiger, Arduino, HTML/CSS, Angular, Node.js, Django, Flask, Spring Boot

Development Tools: Linux/Unix, Visual Studio Code, IntelliJ IDEA CE, Unity, Swift, Flutter, Arduino Uno

Core Competencies: Code Testing & Debugging, Front-End Development, Object-Oriented Programming, Database Management, Game Development

Security & Performance

Cybersecurity Fundamentals: Secure coding practices, authentication, and encryption

Performance Optimization: Profiling tools, caching strategies, and loading balancing.

Clubs & Activities

National Society of Black Engineers (NSBE) | Coalition of Black Women in Engineering (CBWE)

Sisters of Strength and Knowledge (SSK) | LSU National Council of Negro Women

Camp 8 Horse Riders | K-pop Club | Family Youth & Service Center